Confidence Calibration under Ambiguous Ground Truth

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Abstract

Confidence calibration implicitly assumes a unique ground-truth label per input. This assumption breaks down in settings such as medical diagnosis, natural language inference, and perceptual tasks, where multiple annotators may genuinely disagree. We show that post-hoc calibrators fitted on majority-voted labels can appear well-calibrated against voted labels yet remain substantially miscalibrated against labels drawn from the underlying ambiguous label distribution reflected by human disagreement. This failure is structural, affecting classic calibration methods such as Temperature Scaling, Platt scaling, and Histogram Binning. We prove that Temperature Scaling is biased toward lower temperatures than this ambiguity-aware target distribution requires, and that the resulting true-label error increases with annotation entropy. To address this failure, we propose a family of ambiguity-aware post-hoc calibrators that optimize proper scoring rules against the full label distribution, among which **Dirichlet-Soft** — a full $K \times K$ affine calibrator fitted against the annotator distribution — consistently achieves the best distributional fit. For the more common case where the calibration set contains only one-hot voted labels, we propose **Label-Smooth Temperature Scaling (LS-TS)**, which constructs pseudo-soft targets from the model's own confidence without any additional annotations. Experiments on CIFAR-10H, ChaosNLI, ISIC 2019, and DermaMNIST, spanning eight architectures across vision and language, show that Dirichlet-Soft reduces true-label ECE by 48–91% relative to standard Temperature Scaling and achieves the best Brier score and NLL across all benchmarks, while LS-TS reduces true-label ECE by 13–77% without requiring annotator distributions.

1 Introduction

A well-calibrated classifier $f : \mathcal{X} \to \Delta^K$ produces probability vectors $\hat{p}(x) = f(x)$ such that stated confidence matches empirical accuracy: when $\hat{p}_{\hat{c}}(x) = p$ for the predicted class $\hat{c} = \arg\max_k \hat{p}_k(x)$, the true probability of $\hat{c}$ being correct is exactly p. Formally,

$$\mathbb{P}\big(\hat{Y} = Y \mid \hat{p}_{\hat{Y}}(X) = p\big) = p, \quad \forall\, p \in [0, 1], \tag{1}$$

where Δ^K denotes the K-dimensional probability simplex. Calibration is critical for safe deployment in medicine [Kompa et al., 2021], autonomous driving, and other high-stakes settings.

The hidden assumption. Equation (1) implicitly assumes a unique ground-truth label Y for every input x. In practice this assumption frequently fails: a skin lesion may be legitimately classified as malignant or benign by different dermatologists; a natural-language premise may or may not entail a hypothesis depending on pragmatic context; low-resolution objects can be categorically ambiguous. In such settings the correct target is not a single label but a *distribution over labels*, $\pi(\cdot \mid x) \in \Delta^K$, representing the underlying ambiguous label distribution induced by genuine annotator disagreement.

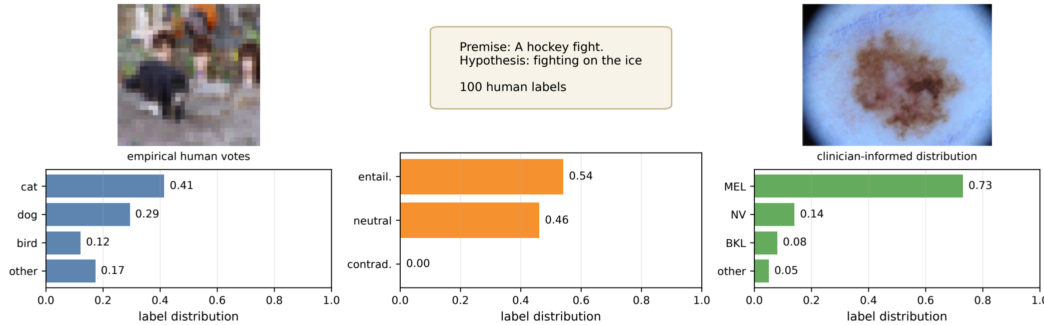

Figure 1: **Left:** a CIFAR-10H image with dispersed human votes (cat/dog/bird), illustrating perceptual ambiguity in low-resolution vision. **Middle:** a ChaosNLI premise–hypothesis pair with split entailment/neutral judgments, illustrating semantic ambiguity. **Right:** an ISIC 2019 melanoma image paired with the clinician-informed label distribution used in our medical experiments, showing clinically plausible MEL/NV confusion. CIFAR-10H and ChaosNLI distributions are empirical human label distributions; the ISIC panel uses the dermatologist confusion model described in Appendix J.

The standard practice and its failure. Figure 1 illustrates the core issue: a single input can support multiple plausible labels, and the full label distribution is the object that should be calibrated. Standard practice nevertheless aggregates annotations by majority vote to obtain a *voted label* y^* and calibrates against it, and in many applications this one-hot voted label is all that is available in the calibration set. We show that this can yield low ECE against voted labels ($\mathrm{ECE}_{\mathrm{voted}}$) while leaving substantial miscalibration against true labels drawn from the ambiguity-aware label distribution $\pi(\cdot \mid x)$ ($\mathrm{ECE}_{\mathrm{true}}$, following Stutz et al. [2023]). The mechanism is straightforward: ambiguous examples appear underconfident relative to the one-hot voted target, so Temperature Scaling selects a lower T than the ambiguity-aware label distribution requires, amplifying overconfidence. In a controlled experiment (Section 4), Temperature Scaling, Platt scaling, and Histogram Binning all fail to correct this mismatch, because the root cause is the voted-label target, not method capacity.

Contributions.

1. **Problem and theory** (Sections 3–5): we formalize true-label calibration under ambiguous labels and show that TS is biased toward overly low temperatures, with true-label miscalibration becoming more severe as annotation entropy increases.
2. **Methods** (Section 6): we propose ambiguity-aware post-hoc calibrators — notably **Dirichlet-Soft**, which consistently achieves the best distributional fit — for settings with annotator distributions, and **LS-TS** for the more realistic setting where the calibration set contains only one-hot voted labels.
3. **Evidence** (Sections 4 and 7): a controlled toy example and four real benchmarks show that standard calibrators can worsen true-label calibration, while Dirichlet-Soft reduces $\mathrm{ECE}_{\mathrm{true}}$ by 48–91% and LS-TS by 13–77% consistently across vision and language.

2 Related Work

Confidence calibration. Guo et al. [2017] showed that modern networks are miscalibrated and that TS is an effective post-hoc remedy. Isotonic regression [Zadrozny and Elkan, 2002], Platt scaling [Platt, 1999], and Dirichlet calibration [Kull et al., 2019] extend this to multiclass settings, all assuming a unique ground-truth label. Minderer et al. [2021] showed ViTs are better calibrated than CNNs; Bai et al. [2021] revisit the sufficiency of TS. All share the voted-label assumption; we show this assumption, not method capacity, is the limiting factor. Label smoothing [Szegedy et al., 2016, Müller et al., 2019] applies uniform smoothed targets during training; we instead use annotation-informed label distributions for post-hoc calibration requiring no retraining.

Ambiguous ground truth. Dense annotation datasets [Peterson et al., 2019, Nie et al., 2020] and multi-annotator learning [Uma et al., 2021, Rodrigues and Pereira, 2018] have received growing attention. Baan et al. [2022] show ECE is theoretically ill-defined when humans genuinely disagree and propose instance-level alternatives; our work complements this by providing methods that improve true-label calibration under that setting. Gordon et al. [2022] and Plank [2022] highlight epistemological problems with annotation aggregation in NLP. Khurana et al. [2024] use disagreement as an abstention signal; Singh et al. [2025] show training on soft labels preserves epistemic uncertainty, a training-time complement to our post-hoc approach.

Conformal prediction under ambiguous ground truth. Stutz et al. [2023] show that conformal sets calibrated on voted labels undercover the true annotator distribution, the analogous failure for set prediction that our work addresses for point prediction. Calibration under covariate shift [Park et al., 2020, Wang et al., 2020] is orthogonal to our label-ambiguity setting.

3 Problem Formulation

3.1 Setup and Notation

Let $\mathcal{Y} = \{1, \ldots, K\}$ be the label space. A classifier $f : \mathcal{X} \to \Delta^K$ outputs $\hat{p}(x) = \mathrm{softmax}(z(x))$ from pre-softmax logits $z(x) \in \mathbb{R}^K$. The **annotator distribution** $\pi(\cdot \mid x) \in \Delta^K$ denotes the underlying ambiguous label distribution for input x and is estimated from m annotations as $\hat{\pi}_k(x) = \frac{1}{m} \sum_j \mathbf{1}[a_j = k]$. The **voted label** is $y^* = \arg\max_k \hat{\pi}_k(x)$. Standard calibration uses one-hot voted labels. For example, Temperature Scaling [Guo et al., 2017] minimizes

$$\mathcal{L}_{\mathrm{TS}}(T) = -\frac{1}{n} \sum_{i=1}^{n} \log \mathrm{softmax}(z_i/T)_{y_i^*}. \tag{2}$$

3.2 True-Label Calibration

Following Stutz et al. [2023], we distinguish the voted label y^* from the **true label** $\tilde{y} \sim \pi(\cdot \mid x)$, drawn from the underlying ambiguous label distribution.

Definition 1 (True-Label Calibration)**.** *f is **true-label calibrated** if for all $p \in [0, 1]$: $\mathbb{P}(\tilde{Y} = \hat{c}(X) \mid \hat{p}_{\hat{c}}(X) = p) = p$, where $\hat{c}(x) = \arg\max_k \hat{p}_k(x)$ and $\tilde{Y} \sim \pi(\cdot \mid X)$.*

The **voted-label ECE** $\mathrm{ECE}_{\text{voted}}$ is the standard ECE against y^*. The **true-label ECE** $\mathrm{ECE}_{\text{true}}$ averages ECE over 100 draws $\tilde{y}_i \sim \pi(\cdot \mid x_i)$ ($B = 15$ equal-width bins). Throughout the paper we compare these two views directly, since ambiguous examples can look well-calibrated under voted labels while remaining overconfident under the underlying ambiguous label distribution.

4 Motivating Example

Setup. We construct a 3-class 2D Gaussian dataset with three clusters: $x \sim \mathcal{N}((-3.2, 1.1), \mathrm{diag}(0.60, 0.45))$ for class 0 with $\pi = [1, 0, 0]$; $x \sim \mathcal{N}((0, 0), \mathrm{diag}(1.15, 0.75))$ for the ambiguous cluster with $\pi = [0, 0.70, 0.30]$; and $x \sim \mathcal{N}((3.2, -1.1), \mathrm{diag}(0.60, 0.45))$ for class 2 with $\pi = [0, 0, 1]$. The voted label for the middle cluster is always class 1 because $0.70 > 0.30$. A 2-layer MLP (64 hidden units per layer) is trained on voted labels, and all calibrators are fitted on a held-out calibration split using the same voted labels.

Results. Figure 2 shows that this mismatch is a structural consequence of the voted-label target. TS reduces $\mathrm{ECE}_{\text{voted}}$ from 3.25% to 1.11%, but *increases* $\mathrm{ECE}_{\text{true}}$ from 6.26% to 9.64%. Platt scaling behaves similarly ($\mathrm{ECE}_{\text{voted}} = 0.68\%$, $\mathrm{ECE}_{\text{true}} = 9.70\%$), and Histogram Binning also lowers voted-label ECE while worsening true-label ECE ($1.32\%/9.65\%$). Figure 2**(f)** further shows that this residual error is concentrated in the ambiguous cluster for all three traditional methods. TS sets $T = 0.62 < 1$, boosting confidence toward the voted target and thereby pushing the model *further* from the true ambiguous label distribution. The failure is therefore not specific to TS: standard post-hoc methods are useful for $\mathrm{ECE}_{\text{voted}}$, but the voted-label target makes them ineffective for $\mathrm{ECE}_{\text{true}}$.

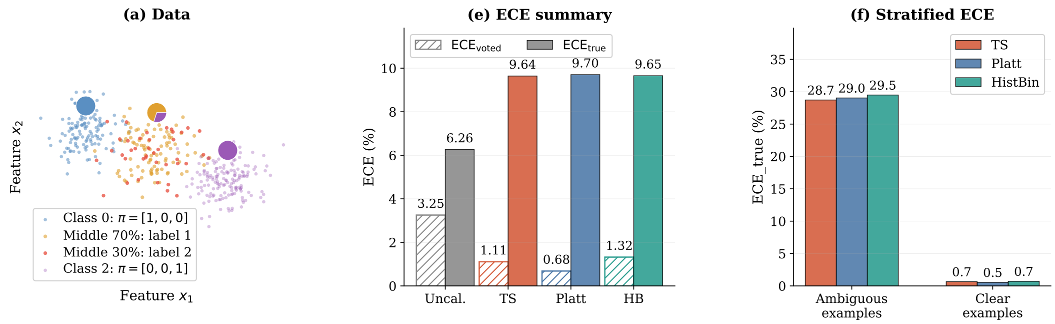

Figure 2: **Motivating example: all standard calibration methods fail.** **(a)** Toy dataset generation: only the middle Gaussian cluster is ambiguous, with $\pi = [0, 0.70, 0.30]$; orange points are drawn as label 1 (70%) and red points as label 2 (30%), while all receive the same voted label 1. **(e)** Summary of the toy results: every traditional calibrator lowers $\text{ECE}_{\text{voted}}$ but *increases* ECE_{true}, so voted-label evaluation masks the failure. **(f)** Stratified ECE_{true} for TS, Platt scaling, and Histogram Binning: the residual error is concentrated in ambiguous examples for all three traditional methods.

5 Theory: Why Standard Calibration Fails

We now formalize two complementary aspects of the failure of voted-label calibration under ambiguity: the *direction* of the TS bias (Proposition 1) and its *scaling* with label ambiguity (Proposition 2).

Proposition 1 (Direction of TS Bias)**.** *Consider a calibration set containing an ambiguous cluster in which the voted label y^* equals the majority class for every example, but the true annotator probability of y^* is $\pi_{y^*}(x) = q < 1$. If the model is already reasonably accurate on this cluster ($\text{softmax}(z_i)_{y^*} > q$), then TS minimizing $\mathcal{L}_{TS}$ finds $T^* < 1$ (increasing model confidence), whereas ambiguity-aware calibration requires $T^* > 1$ (reducing confidence to q).*

Proof sketch. On the ambiguous cluster, all voted labels are y^*, so the voted-label loss (2) is minimized by pushing $\text{softmax}(z_i/T)_{y^*} \to 1$, i.e. $T \to 0$. Hence $\partial\mathcal{L}_{\text{TS}}/\partial T > 0$ and the optimal $T^* < 1$. Ambiguity-aware calibration instead minimizes $\mathcal{L}_{\text{SLTS}}$ with target $q < 1$; the minimum requires $\text{softmax}(z_i/T^*)_{y^*} \approx q$, necessitating $T^* > 1$ whenever $\text{softmax}(z_i)_{y^*} > q$. Full proof in Appendix C. □

Proposition 2 (True-Label Miscalibration and Annotation Entropy)**.** *Let* $H(x) = -\sum_k \pi_k(x) \log \pi_k(x)$ *denote the annotation entropy of x. Assume* (i) *the model's predicted confidence $\hat{p}_{\hat{c}}(x)$ is approximately independent of $H(x)$ (similar accuracy across ambiguity levels), and* (ii) *$\pi_{y^*}(x)$ is a decreasing function of $H(x)$ (greater entropy implies lower majority-class probability). Then the expected true-label calibration error of voted-label-calibrated predictions is non-decreasing in $H(x)$.*

Proof sketch. For unambiguous examples ($H(x) = 0$), $\pi(\cdot \mid x)$ is one-hot, so true-label and voted-label evaluation coincide. As $H(x)$ increases, $\pi_{y^*}(x)$ decreases below 1. The true-label calibration error for the predicted class is $|\hat{p}_{\hat{c}}(x) - \pi_{\hat{c}}(x)|$; since the model is trained against the voted label (targeting ≈ 1), $\hat{p}_{\hat{c}}(x) > \pi_{y^*}(x)$, and this mismatch grows as $\pi_{y^*}(x)$ falls. Hence the true-label calibration error increases with $H(x)$. □

6 Methods

All methods in this paper are *post-hoc*: they operate on cached logits from a fixed pre-trained model and require only a calibration set equipped with annotator distributions or individual annotations. No retraining or architectural modification is needed. Training-time approaches such as soft-label training [Szegedy et al., 2016, Müller et al., 2019], mixup [Thulasidasan et al., 2019], and multi-annotator learning [Uma et al., 2021, Rodrigues and Pereira, 2018] can also improve calibration under ambiguity, but they require access to the training pipeline and are orthogonal to post-hoc calibration; we do not compare against them here.

6.1 Ambiguity-Aware Calibration Methods

Soft-Label Temperature Scaling (SLTS). SLTS minimizes the cross-entropy between the calibrated output and the empirical label distribution:

$$\begin{aligned}\mathcal{L}_{\mathrm{SLTS}}(T) &= -\frac{1}{n}\sum_{i=1}^{n}\sum_{k=1}^{K}\hat{\pi}_k(x_i)\,\log\mathrm{softmax}(z_i/T)_k \\ &= \frac{1}{n}\sum_{i=1}^{n}\mathrm{KL}\big(\hat{\pi}(x_i)\,\|\,\mathrm{softmax}(z_i/T)\big) + \text{const.}\end{aligned} \tag{3}$$

SLTS uses the same single-parameter family as TS but with a target that correctly reflects the annotator distribution.

Proposition 3 (Proper scoring rule)**.** *$\mathcal{L}_{SLTS}$ is a proper scoring rule with respect to $\hat{\pi}$: it is uniquely minimized (over all distributions, not just the one-parameter TS family) when $\mathrm{softmax}(z/T) = \hat{\pi}(x)$ for every x.*

Vector Scaling (VS). Extends SLTS with per-class temperatures $\mathbf{T} = (T_1, \ldots, T_K)$, accommodating datasets where some classes are systematically more ambiguous than others.

Distributional Isotonic Regression (IR-Soft). Fits a monotone mapping from predicted confidence to mean annotator accuracy via the Pool Adjacent Violators Algorithm (PAVA), using the empirical label distribution as target.

SoftPlatt. The diagonal affine transform $W \cdot z + b$ (same structure as Platt) fitted against the annotator distribution via KL divergence. This isolates the contribution of the distributional target from any architectural difference.

Dirichlet-Soft. The full $K \times K$ affine $Wz + b$ fitted by minimizing $\mathrm{KL}(\hat{\pi}(x)\|\,\mathrm{softmax}(Wz + b))$ with ODIR regularization ($\lambda = 10^{-3}$) [Kull et al., 2019]. This is the most expressive parametric ambiguity-aware calibrator.

6.2 Annotation-Free Calibration

The methods above require annotator distributions $\hat{\pi}(x_i)$ at calibration time. When only voted labels y_i^* are available, we propose **Label-Smooth Temperature Scaling (LS-TS)**, which constructs a pseudo-target distribution from the model's own predictions without any additional annotations.

Let $\bar{\varepsilon} = \frac{1}{n}\sum_{i=1}^{n}\big(1 - \hat{p}_{y_i^*}(x_i)\big)$ be the mean complement of the model's voted-class confidence on the calibration set. Define the pseudo-target distribution

$$\tilde{\pi}_i^{\mathrm{LS}} = (1-\bar{\varepsilon})\,e_{y_i^*} + \tfrac{\bar{\varepsilon}}{K}\,\mathbf{1}, \tag{4}$$

and minimize $\mathcal{L}_{\mathrm{SLTS}}$ (Eq. 3) with these targets. The global smoothing weight $\bar{\varepsilon}$ is a data-driven estimate of average annotator disagreement: if the model is 80% confident on the voted class, the remaining 20% is spread uniformly across all K classes. The result is $T^*_{\mathrm{LS}} > T^*_{\mathrm{TS}}$ whenever $\bar{\varepsilon} > 0$, moving the calibrator in the correct direction even without annotator data.

Theoretical interpretation. The smoothing weight $\bar{\varepsilon}$ satisfies a moment-matching fixed point: the expected pseudo-label mass on the voted class equals the average model confidence, $\mathbb{E}[\tilde{\pi}^{\mathrm{LS}}_{y^*}] = \mathbb{E}[\hat{p}_{y^*}]$. Algorithmically, LS-TS can be viewed as a single E-step of an EM algorithm in which the latent annotator distribution is estimated from the current (pre-calibration) model, followed by one M-step that optimises T given those pseudo-labels. This connection to self-distillation [Hinton et al., 2015] explains the self-referential nature of the approach: the uncalibrated model both supplies the pseudo-targets and is then calibrated against them.

7 Experiments

We evaluate on four benchmarks with multi-annotator data, each with two backbones: CIFAR-10H (ResNet-50, ViT-B/16), ChaosNLI (RoBERTa-Large, DeBERTa-v3), ISIC 2019 (EfficientNet-B4, ViT-S/16), and DermaMNIST (ResNet-18, ViT-S/16).

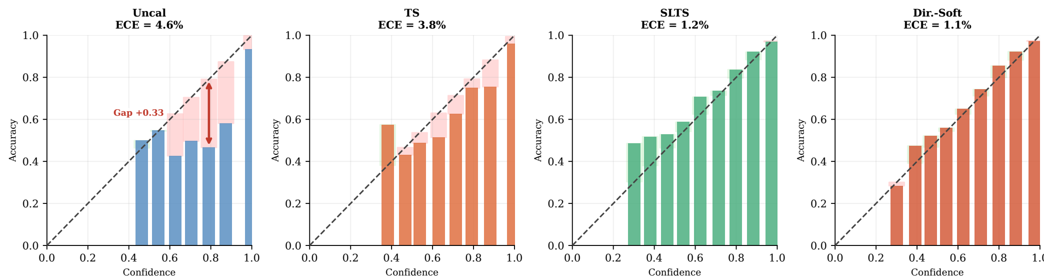

Figure 3: **Reliability diagrams for four representative methods — CIFAR-10H ResNet-50 ($\mathbf{ECE_{true}}$).** Red shading indicates overconfidence; green indicates underconfidence. Uncal and TS remain overconfident; SLTS substantially corrects this with a single temperature parameter; Dirichlet-Soft reduces the residual gap further. Reliability diagrams for all remaining methods are provided in Appendix H.

Baselines. We compare against (i) **Uncalibrated**: raw softmax; (ii) **TS**: Temperature Scaling on voted labels [Guo et al., 2017]; (iii) **Platt (PS)**: per-class weight and bias on logits, calibrated against voted labels [Platt, 1999, Kull et al., 2019]; (iv) **Dirichlet-Hard**: Dirichlet calibration [Kull et al., 2019] with the full $K \times K$ weight matrix and bias, calibrated against voted labels. Dirichlet-Hard uses the *same architecture* as our Dirichlet-Soft but with voted-label targets; the contrast isolates the effect of the calibration target from model capacity.

Metrics. All metrics are evaluated against true labels $\tilde{y} \sim \pi(\cdot \mid x)$: $\mathrm{ECE}_{\text{true}}$ (averaged over 100 draws; $B = 15$ equal-width bins), adaptive ECE (aECE_t; equal-mass bins, same draws), classwise ECE (cwECE_t; per-class ECE averaged over K classes [Kull et al., 2019]), Brier score (Br_t), and NLL (NLL_t). We emphasize $\mathrm{ECE}_{\text{true}}$ because it mirrors the evaluation protocol a practitioner faces when only one label is observed at test time; the Brier score and NLL directly measure distributional alignment with $\hat{\pi}$ and serve as complementary distributional metrics. We abbreviate $\mathrm{ECE}_{\text{true}}$, aECE, cwECE, Br, NLL throughout.

7.1 CIFAR-10H: Image Classification with Human Label Noise

Dataset and models. CIFAR-10H [Peterson et al., 2019] provides ≈51 human annotations per image for the 10 000-image CIFAR-10 test set, directly exposing perceptual ambiguity in low-resolution natural images through repeated human labeling. The test set is split into calibration ($n = 5{,}000$) and evaluation (5,000) by stratified sampling (seed 42). We evaluate two architectures: **ResNet-50** [He et al., 2016] pretrained on ImageNet-1k, fine-tuned for 30 epochs (AdamW, cosine annealing), achieving 97.3% top-1 accuracy. **ViT-B/16** [Dosovitskiy et al., 2021] pretrained on ImageNet-21k, fine-tuned with the same protocol, achieving 98.1% top-1 accuracy.

Results. Both CIFAR-10H backbones show the same pattern: hard-label baselines still leave 4–5% $\mathrm{ECE}_{\text{true}}$, while ambiguity-aware methods reduce it to below 2%. Dirichlet-Soft achieves the best Brier score and NLL on both architectures (0.109/0.271 on ResNet-50; 0.100/0.255 on ViT-B/16), confirming that expressive distribution-aware calibrators provide the best distributional fit. LS-TS, using only voted labels, already reduces ECE from 4.29% to 1.57% (ResNet-50) and from 4.48% to 2.37% (ViT-B/16), competitive with fully ambiguity-aware methods.

7.2 ChaosNLI: Natural Language Inference

We next evaluate on **ChaosNLI** [Nie et al., 2020]: 100 human annotations per example for the combined SNLI+MNLI subset ($n = 3{,}113$), where disagreement reflects genuine semantic ambiguity rather than annotation noise. The data are split 50/50 stratified by voted label. We evaluate **RoBERTa-Large** [Liu et al., 2019] and **DeBERTa-v3-base** [He et al., 2023], both fine-tuned on MNLI.

Results. The same pattern holds in NLI: hard-label baselines leave 10–12% ECE, while Dirichlet-Soft achieves the best overall distributional fit (Brier 0.510/0.509; NLL 0.839/0.836 on RoBERTa-L/DeBERTa-v3). LS-TS reduces ECE from 10.55% to 4.16% (RoBERTa-L) and from 11.63% to

Table 1: **CIFAR-10H and ChaosNLI results.** All metrics evaluated against true (sampled) labels. Best per architecture in **bold**.

	ResNet-50						**ViT-B/16**					
Method	T	ECE↓	aECE↓	cwECE↓	Br↓	NLL↓	T	ECE↓	aECE↓	cwECE↓	Br↓	NLL↓
CIFAR-10H *(≈51 annotations per image, K=10)*												
Voted-label baselines												
Uncalibrated	—	4.97	5.71	1.09	0.120	0.692	—	4.99	5.36	1.06	0.111	0.678
TS	2.03	4.29	4.25	0.91	0.112	0.363	2.04	4.48	4.40	0.93	0.106	0.346
Platt	—	4.29	4.23	0.91	0.112	0.372	—	4.54	4.38	0.93	0.105	0.363
Dirichlet-Hard	—	4.46	4.44	0.95	0.114	0.395	—	4.70	4.62	0.97	0.107	0.394
Ambiguity-aware (ours)												
SLTS	3.18	1.51	1.39	0.45	0.110	0.293	3.07	0.85	0.56	0.39	0.102	0.278
SoftPlatt	—	1.52	1.31	**0.35**	0.110	0.288	—	0.88	0.47	**0.24**	0.101	0.272
VS	—	1.35	1.26	0.39	**0.109**	0.289	—	0.92	0.50	0.32	0.101	0.274
IR-Soft	—	**0.72**	**0.83**	0.45	0.115	0.340	—	0.91	0.61	0.43	0.105	0.321
Dirichlet-Soft	—	1.25	1.06	**0.35**	**0.109**	**0.271**	—	**0.72**	**0.41**	0.25	**0.100**	**0.255**
Annotation-free (ours)												
LS-TS	3.08	1.57	1.31	0.46	0.109	0.293	2.76	2.37	2.21	0.53	0.103	0.285
	RoBERTa-Large						**DeBERTa-v3**					
Method	T	ECE↓	aECE↓	cwECE↓	Br↓	NLL↓	T	ECE↓	aECE↓	cwECE↓	Br↓	NLL↓
ChaosNLI *(SNLI+MNLI, 100 annotations per example, K=3)*												
Voted-label baselines												
Uncalibrated	—	27.79	27.77	18.61	0.650	1.384	—	35.97	35.92	24.02	0.743	2.148
TS	2.42	10.55	10.38	7.30	0.537	0.886	3.81	11.63	11.57	8.48	0.549	0.900
Platt	—	11.16	10.92	7.28	0.530	0.875	—	12.43	12.36	8.43	0.539	0.896
Dirichlet-Hard	—	11.55	11.40	7.64	0.535	0.889	—	12.69	12.63	8.84	0.539	0.895
Ambiguity-aware (ours)												
SLTS	3.41	3.22	2.59	4.62	0.520	0.862	5.28	3.45	3.36	6.53	0.529	0.877
SoftPlatt	—	2.66	2.32	2.43	**0.509**	**0.839**	—	3.17	2.92	3.18	0.511	0.841
VS	—	3.46	3.07	5.13	0.518	0.857	—	3.91	3.35	6.46	0.525	0.869
IR-Soft	—	2.65	2.59	6.24	0.549	0.934	—	**2.15**	3.96	7.20	0.554	0.942
Dirichlet-Soft	—	**2.57**	**1.71**	**2.03**	0.510	**0.839**	—	3.19	**2.71**	**2.95**	**0.509**	**0.836**
Annotation-free (ours)												
LS-TS	4.40	4.16	3.72	5.61	0.522	0.873	6.15	2.65	2.99	6.32	0.529	0.881

2.65% (DeBERTa-v3) using only voted labels, outperforming all hard-label baselines by a wide margin.

7.3 ISIC 2019: Skin Disease Diagnosis

We evaluate on **ISIC 2019** [Codella et al., 2019, Combalia et al., 2019] (8 skin conditions, ≈25,000 images), a clinically realistic differential-diagnosis task in which several lesion categories have overlapping visual morphology. We use $K = 9$ synthetic dermatologist annotations per image, generated from a clinically calibrated 8×8 confusion matrix (Appendix J) matching Liu et al. [2020]: mean diagonal agreement 75%. We evaluate **EfficientNet-B4** [Tan and Le, 2019] and **ViT-S/16** [Dosovitskiy et al., 2021] on a stratified 70/15/15 split.

ISIC 2019 results. The medical setting shows the same failure mode for standard calibrators and the same benefit from ambiguity-aware targets. Dirichlet-Soft again achieves the best Brier and NLL on both backbones (0.518/1.084 on ENet-B4; 0.563/1.175 on ViT-S/16), while IR-Soft achieves the lowest ECE. LS-TS halves the ECE of TS on ENet-B4 (18.54%→9.34%) using only voted labels; the improvement is smaller on ViT-S/16 (16.59%→15.49%), highlighting that model confidence is not always a reliable proxy for annotation ambiguity.

7.4 DermaMNIST: Supplementary Skin Disease Experiment

We additionally evaluate on **DermaMNIST** [Yang et al., 2023] (7-class, derived from HAM10000 [Tschandl et al., 2018]; $K = 5$ synthetic annotators; overall agreement ≈ 64.7%) with two backbones: ResNet-18 and ViT-S/16. Results in Table 2 show that TS leaves ECE = 23.1% for ResNet-18

Table 2: **ISIC 2019 and DermaMNIST results.** All metrics evaluated against true (sampled) labels. Best per architecture in **bold**.

	EfficientNet-B4						ViT-S/16					
Method	T	ECE↓	aECE↓	cwECE↓	Br↓	NLL↓	T	ECE↓	aECE↓	cwECE↓	Br↓	NLL↓
ISIC 2019 *(K=9 synthetic annotators, mean diagonal agreement* 75%*)*												
Voted-label baselines												
Uncalibrated	—	25.76	25.84	6.67	0.600	2.710	—	23.31	23.33	6.55	0.651	1.978
TS	1.79	18.54	18.49	5.02	0.560	1.653	1.42	16.59	16.60	5.23	0.617	1.563
Platt	—	19.89	19.81	5.24	0.562	1.659	—	17.19	17.17	4.74	0.597	1.583
Dirichlet-Hard	—	20.36	20.32	5.33	0.563	1.674	—	17.76	17.72	4.86	0.600	1.597
Ambiguity-aware (ours)												
SLTS	4.46	9.71	9.74	2.92	0.528	1.146	3.15	6.96	6.68	3.29	0.586	1.233
SoftPlatt	—	9.25	9.31	2.23	0.523	1.108	—	8.05	7.90	2.05	0.565	1.190
VS	—	9.03	9.11	2.42	0.524	1.121	—	8.50	8.59	2.24	0.567	1.198
IR-Soft	—	**1.75**	**2.43**	4.28	0.538	1.284	—	**1.73**	**1.59**	5.78	0.621	1.466
Dirichlet-Soft	—	7.50	7.48	**2.00**	**0.518**	**1.084**	—	6.85	6.80	**1.77**	**0.563**	**1.175**
Annotation-free (ours)												
LS-TS	4.30	9.34	9.46	2.98	0.528	1.149	3.98	15.49	15.47	4.52	0.619	1.303
	ResNet-18						ViT-S/16					
Method	T	ECE↓	aECE↓	cwECE↓	Br↓	NLL↓	T	ECE↓	aECE↓	cwECE↓	Br↓	NLL↓
DermaMNIST *(K=5 synthetic annotators, overall agreement* 64.7%*)*												
Voted-label baselines												
Uncalibrated	—	34.35	34.26	10.15	0.767	2.689	—	37.12	37.10	10.82	0.786	3.766
TS	1.86	23.05	22.82	6.97	0.686	1.654	2.60	24.36	24.26	7.29	0.683	1.664
Platt	—	23.64	23.56	7.08	0.682	1.667	—	23.82	23.59	7.10	0.675	1.681
Dirichlet-Hard	—	24.58	24.46	7.33	0.690	1.771	—	24.84	24.68	7.49	0.682	1.897
Ambiguity-aware (ours)												
SLTS	3.51	4.61	4.44	3.56	0.622	1.376	5.11	3.58	3.06	3.37	0.610	1.350
SoftPlatt	—	4.19	3.55	1.61	0.612	1.312	—	3.56	2.93	1.29	**0.601**	1.289
VS	—	4.45	3.83	3.23	0.621	1.363	—	4.03	3.71	2.84	0.611	1.331
IR-Soft	—	**2.15**	**2.34**	4.55	0.634	1.440	—	3.79	3.73	4.59	0.628	1.431
Dirichlet-Soft	—	3.55	3.36	**1.22**	**0.610**	**1.305**	—	**3.32**	**2.87**	**1.14**	**0.601**	**1.286**
Annotation-free (ours)												
LS-TS	3.03	5.05	4.82	3.79	0.630	1.394	4.19	7.36	7.06	3.68	0.615	1.362

while ambiguity-aware methods reduce it substantially: SLTS achieves 4.6% (80% reduction), IR-Soft achieves 2.2% (91% reduction), and Dirichlet-Soft achieves 3.6% while attaining the best cwECE (1.2%), Brier (0.610), and NLL (1.305). ViT-S/16 shows a similar pattern (SLTS: 3.6%; Dirichlet-Soft achieves the best ECE, Brier, and NLL: 3.3%/0.601/1.286). Both backbones show $T_{\text{SLTS}} > T_{\text{TS}} > 1$: SLTS requires substantially higher temperature to match the more diffuse annotation distribution.

7.5 Reliability Diagrams

Figure 3 shows reliability diagrams for four representative methods on CIFAR-10H (ResNet-50). The diagrams are evaluated against the true label distribution (ECE_{true}). Uncal and TS are systematically overconfident: bars fall well below the diagonal at high confidence, leaving a large red gap. SLTS corrects the overconfidence with a scalar temperature adjustment, aligning most bins close to the diagonal. Dirichlet-Soft achieves near-perfect calibration by combining the expressive $K \times K$ affine transform with the correct distributional target. Corresponding diagrams for ChaosNLI (RoBERTa-Large) are provided in Appendix H, along with full 3×5 panels covering all 13 methods.

8 Discussion

Sacrificing voted-label calibration. A natural question is whether ambiguity-aware methods degrade $\text{ECE}_{\text{voted}}$. This is expected behaviour: voted-label calibration is epistemically unreliable when annotators genuinely disagree, as the voted label overstates the majority-class probability.

Practitioners who must satisfy a hard $\text{ECE}_{\text{voted}}$ constraint can apply TS on top of Dirichlet-Soft or SLTS outputs.

Practical guidelines. Use **Dirichlet-Soft** as the default when annotator distributions are available — it achieves the best Brier score and NLL across all benchmarks while maintaining competitive ECE. Use **SLTS** when a single-parameter method is preferred for simplicity; **IR-Soft** when the lowest ECE is the sole criterion; and **LS-TS** when the calibration set contains only one-hot voted labels.

Robustness. Appendix G confirms that the ranking of methods is stable across random seeds (std ≤ 0.54 pp) and Appendix F shows that ambiguity-aware methods converge with as few as 5–10% of the calibration set.

Limitations. Full ambiguity-aware calibration (including Dirichlet-Soft) requires multi-annotator labels at calibration time. When only voted labels are available, LS-TS remains useful but does not fully match Dirichlet-Soft, particularly on datasets where model confidence diverges from annotation ambiguity (e.g., ISIC 2019 ViT-S/16). ISIC 2019 and DermaMNIST rely on synthetic annotators derived from clinician-reported agreement, so real multi-reader medical calibration sets would provide stronger validation.

9 Conclusion

We formalized the problem of confidence calibration under ambiguous ground truth and showed that standard post-hoc calibrators fitted on voted labels are systematically miscalibrated against the underlying ambiguous label distribution. Our flagship ambiguity-aware method, **Dirichlet-Soft**, requires no retraining and achieves the best Brier score and NLL across all four benchmarks and eight architectures spanning vision and language, reducing true-label ECE by 48–91% relative to Temperature Scaling. The key insight is that the calibration target — not method capacity — is the limiting factor: Dirichlet calibration with voted-label targets (Dirichlet-Hard) consistently *underperforms* simple TS, while the same architecture with soft-label targets (Dirichlet-Soft) achieves the best overall results. Crucially, when the calibration set contains only one-hot voted labels, our annotation-free method **LS-TS** still improves true-label calibration, reducing ECE by 13–77% using only the model's own confidence as a proxy for annotator disagreement. We recommend Dirichlet-Soft as the default wherever multi-annotator data is available, and LS-TS as a practical post-hoc alternative when only voted labels are available.

Acknowledgements. We thank the anonymous reviewers for their constructive feedback.

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A Dataset Background and Sources of Label Ambiguity

All four benchmarks in this paper are appropriate for studying calibration under ambiguous ground truth, but they exhibit ambiguity in slightly different ways.

CIFAR-10H. Peterson et al. [2019] introduced CIFAR-10H by collecting a full distribution of human labels for each image in the CIFAR-10 test set, explicitly to capture *human perceptual uncertainty*. The ambiguous examples are typically low-resolution images whose visual evidence supports multiple plausible classes for human observers. In this benchmark, the label distribution is therefore directly observed from repeated human annotation, making it a clean testbed for ambiguity-aware calibration.

ChaosNLI. Nie et al. [2020] introduced ChaosNLI to study *collective human opinions* on natural language inference. Each example receives 100 human labels, and the paper reports that substantial disagreement is common rather than exceptional. This ambiguity is semantic rather than perceptual: differences in pragmatic assumptions, underspecification, and sentence interpretation lead reasonable annotators to assign different NLI labels to the same premise–hypothesis pair.

ISIC 2019. ISIC 2019 is a multiclass dermoscopic lesion diagnosis benchmark assembled from several clinical sources, including BCN20000 [Codella et al., 2019, Combalia et al., 2019]. The task itself is ambiguity-prone because several lesion categories share overlapping visual morphology, and the challenge was designed to reflect the more realistic task of differential diagnosis rather than a simpler benign-versus-malignant decision. Because the public challenge release does not provide per-image reader distributions, we model ambiguity using synthetic annotators calibrated to dermatologist confusion patterns and reader agreement reported in Liu et al. [2020]. This preserves the core property relevant to our study: in dermatology, a single image can support multiple clinically plausible labels.

DermaMNIST. DermaMNIST [Yang et al., 2023] is a standardized 28×28 version of the dermatology benchmark derived from HAM10000 [Tschandl et al., 2018]. HAM10000 consists of dermatoscopic images of common pigmented lesions collected in clinical practice, where diagnosis already involves fine-grained distinctions among visually similar categories such as melanoma, nevi, and benign keratoses. DermaMNIST inherits this medical ambiguity from HAM10000 and, because it is aggressively downsampled for lightweight benchmarking, removes additional visual detail that can further increase label uncertainty. As with ISIC 2019, we therefore evaluate calibration against an ambiguity-aware label distribution rather than treating the majority diagnosis as uniquely correct.

B ISIC 2019 Full Results

See Table 2 in the main text.

C Proof of Proposition 1

Let $\mathcal{D}_{\text{amb}} = \{(x_i, y_i^*) : y_i^* = c,\, x_i \in \mathcal{C}\}$ be the subset of calibration examples from the ambiguous cluster $\mathcal{C}$, where the voted label is always class c. Denote $p_i = \mathrm{softmax}(z_i)_c$.

The TS loss on $\mathcal{D}_{\text{amb}}$ is $\mathcal{L}_{\text{TS}}(T; \mathcal{D}_{\text{amb}}) = -|\mathcal{D}_{\text{amb}}|^{-1} \sum_i \log \mathrm{softmax}(z_i/T)_c$. As $T \to 0^+$, $\mathrm{softmax}(z_i/T)_c \to 1$ (assuming $z_{ic} > z_{ik}$ for all $k \neq c$), so $\mathcal{L}_{\text{TS}} \to 0$. Hence $\partial \mathcal{L}_{\text{TS}}/\partial T > 0$ for all $T > 0$: decreasing T always decreases the voted-label loss. The optimal T^* balances contributions from ambiguous and unambiguous examples, but $T^* \leq T^*_{\text{no-amb}}$ (the optimal temperature without the ambiguous cluster). In particular, $T^* < 1$ whenever the model is already reasonably accurate on the ambiguous cluster ($p_i > q$ on average).

The SLTS loss on $\mathcal{D}_{\text{amb}}$ is $\mathcal{L}_{\text{SLTS}}(T; \mathcal{D}_{\text{amb}}) = -|\mathcal{D}_{\text{amb}}|^{-1} \sum_i \sum_k \hat{\pi}_k(x_i) \log \mathrm{softmax}(z_i/T)_k$, with $\hat{\pi}_c(x_i) = q < 1$. The minimum-loss T^* satisfies $\mathrm{softmax}(z_i/T^*)_c \approx q$, requiring $T^* > 1$ whenever $p_i > q$. $\square$

D Proof of Proposition 2

We formalize the argument given in the proof sketch in the main text.

Setup. Let $p = \hat{p}_{\hat{c}}(x)$ denote the model's predicted confidence for the top class $\hat{c}$ and let $q = \pi_{\hat{c}}(x) \in (0, 1]$ denote the true annotator probability for that class. We assume the model is trained against voted labels, so p is approximately constant across examples with the same voted label regardless of their annotation entropy $H(x)$.

Voted-label calibration error contribution. For example x_i in confidence bin B_b (where $\overline{p}(B_b) \approx p$), the per-example voted-label calibration error is $|p - \mathbf{1}[\hat{c} = y^*]|$. Since $\hat{c} = y^*$ for correctly-classified ambiguous examples (the model predicts the majority class), this contribution is approximately $|p-1|$ for examples where the model makes the right hard prediction, independent of $H(x)$.

True-label calibration error contribution. The per-example true-label calibration error is $|p - \pi_{\hat{c}}(x)| = |p-q|$.

Excess true-label error relative to voted-label evaluation. The per-example excess contribution is:

$$\delta(x) = |p-q| - |p - \mathbf{1}[\hat{c} = y^*]|. \tag{5}$$

When the model is correctly calibrated to voted labels ($p \approx \mathbf{1}[\hat{c} = y^*]$ in expectation), the voted contribution is near zero: $|p-1| \approx 0$ for high-accuracy bins and $|p-0| = p$ for error bins. For unambiguous examples ($H(x) = 0$), $q = \pi_{y^*} = 1$, so $\delta(x) = |p-1| - |p-1| = 0$.

Monotonicity. As $H(x)$ increases, $q = \pi_{y^*}(x)$ decreases below 1 (more probability mass shifts to non-majority classes). Specifically, for a K-class problem with uniform annotator distribution at maximum entropy, $q = 1/K$. The true-label contribution $|p-q|$ increases strictly as q decreases from 1 (since $p > q$ for a well-trained model predicting the majority class with confidence exceeding the majority probability). The voted-label contribution $|p - \mathbf{1}[\hat{c} = y^*]|$ is unaffected by $H(x)$. Therefore $\delta(x) = |p-q| - |p - \mathbf{1}[\hat{c} = y^*]|$ is non-decreasing in $H(x)$, and strictly increasing whenever $p > q > 0$.

Aggregate monotonicity. The aggregate excess true-label error relative to voted-label evaluation equals $\sum_b \frac{|B_b|}{n} \sum_{i \in B_b} \delta(x_i)/|B_b|$. Since $\delta(x_i)$ is non-decreasing in $H(x_i)$ and bins contain a mixture of examples, this discrepancy grows with the mean annotation entropy of the test set. $\square$

Remark 1 (Independence assumption)**.** *Assumption (i) (that $\hat{p}_{\hat{c}}(x)$ is approximately independent of $H(x)$) may be violated in practice if harder examples (high $H(x)$) also have lower model confidence. When the two are positively correlated, the per-example discrepancy term $\delta(x)$ grows even faster with $H(x)$ than the proof suggests, so Proposition 2 remains valid and the monotonicity is if anything* conservative.

E Monte Carlo Temperature Scaling (MCTS)

Method. When individual annotation records are available rather than pre-aggregated label distributions, **MCTS** draws S samples $a_{is} \sim \hat{\pi}(x_i)$ per example and minimizes

$$\mathcal{L}_{\text{MCTS}}(T) = -\frac{1}{nS} \sum_{i=1}^{n} \sum_{s=1}^{S} \log \operatorname{softmax}(z_i/T)_{a_{is}}.$$

MCTS is a strictly more general formulation than SLTS: it applies even when only raw annotation samples are stored rather than pre-computed frequency vectors.

Convergence to SLTS. Since $\mathbb{E}_{\hat{y} \sim \hat{\pi}}[\text{CE}(\operatorname{softmax}(z/T), \hat{y})] = \text{KL}(\hat{\pi} \| \operatorname{softmax}(z/T)) + H(\hat{\pi})$, the MCTS objective converges to SLTS in expectation as $S \to \infty$, with variance $\propto 1/S$. Table 3 reports the corresponding sweep on CIFAR-10H (ResNet-50). Even small S already tracks SLTS closely, and the variance shrinks as expected; in practice $S = 20$–50 is sufficient.

F Ablation: Calibration Set Size

We fix the annotation count at the full observed value and vary the fraction of calibration examples used, from 5% to 100% (stratified subsampling by class). Figure 4 shows results for CIFAR-10H (ResNet-50, ViT-B/16) and ChaosNLI (RoBERTa-Large).

Key findings. On CIFAR-10H ResNet-50, TS's ECE fluctuates around 3.9% for all calibration set sizes (5%: 3.4%; 100%: 3.9%), confirming that the bias is structural rather than due to insufficient data. In contrast, SLTS achieves $\approx$1.4% ECE stably from 5% onwards, and IR-Soft and Dirichlet-Soft

Table 3: **MCTS convergence on CIFAR-10H (ResNet-50).** Mean $\pm$ std over 5 seeds; S = MC annotation samples per calibration example.

Method	ECE (%)	T^*
TS (voted labels)	4.29	2.030
MCTS $S = 1$	1.53 ± 0.01	3.174 ± 0.041
MCTS $S = 5$	1.53 ± 0.02	3.166 ± 0.014
MCTS $S = 20$	1.52 ± 0.00	3.187 ± 0.013
MCTS $S = 50$	1.52 ± 0.01	3.185 ± 0.007
MCTS $S = 200$	1.52 ± 0.00	3.178 ± 0.003
SLTS ($S \to \infty$)	1.51	3.180

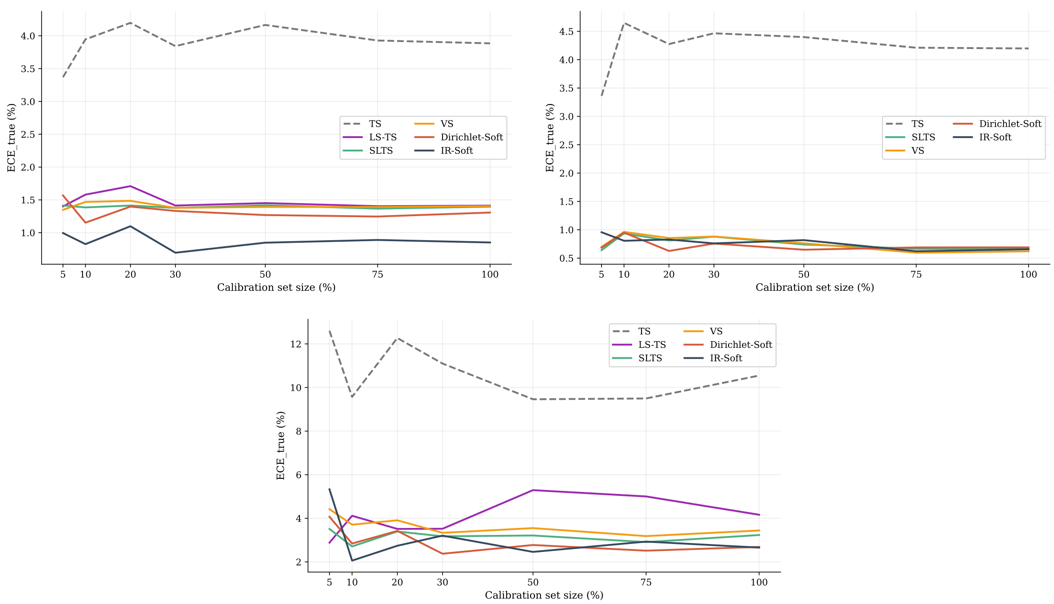

Figure 4: **Calibration-set-size ablation.** ECE_{true} vs. fraction of calibration data used (full annotation count; only the number of calibration examples varies). Top: CIFAR-10H ResNet-50 (left) and ViT-B/16 (right). Bottom: ChaosNLI RoBERTa-Large. TS's ECE does not decrease with more calibration data, confirming that the degradation is a target bias rather than a variance artifact. Ambiguity-aware methods converge quickly and remain stable even at 5–10% of the calibration set.

converge below 1.3% with as few as 10% of the calibration set. The same pattern holds for ChaosNLI: TS fluctuates around 10–13% while SLTS and IR-Soft stabilise near 3% and 2.5%, respectively. This suggests that the annotation distribution (not the number of calibration examples) is the limiting factor for ambiguity-aware methods.

G Statistical Significance: Multi-Seed Analysis

To verify that our main results are not sensitive to the particular calibration/test split (seed 42), we repeat the evaluation with five random seeds (42–46), each producing a different stratified 50/50 split. Table 4 reports mean $\pm$ standard deviation of ECE_{true} across seeds for CIFAR-10H (ResNet-50) and ChaosNLI (RoBERTa-Large).

All methods show small standard deviations (≤ 0.54 percentage points), while the ECE gaps between TS and the ambiguity-aware methods are 2–$9\times$ larger than the standard deviation, confirming that the improvements reported in Tables 1–2 are statistically robust.

Table 4: **Multi-seed stability.** Mean $\pm$ std of $\mathrm{ECE}_{\mathrm{true}}$ (%) over 5 random seeds (42–46), each inducing a different stratified cal/test split. Standard deviations are small relative to the ECE gaps, confirming that the ranking of methods is stable across splits.

Method	CIFAR-10H ResNet-50	ChaosNLI RoBERTa-L
TS	4.25 ± 0.25	11.54 ± 0.54
SLTS	1.61 ± 0.12	3.85 ± 0.34
Dirichlet-Soft	1.37 ± 0.09	2.90 ± 0.14
IR-Soft	0.89 ± 0.12	2.40 ± 0.18

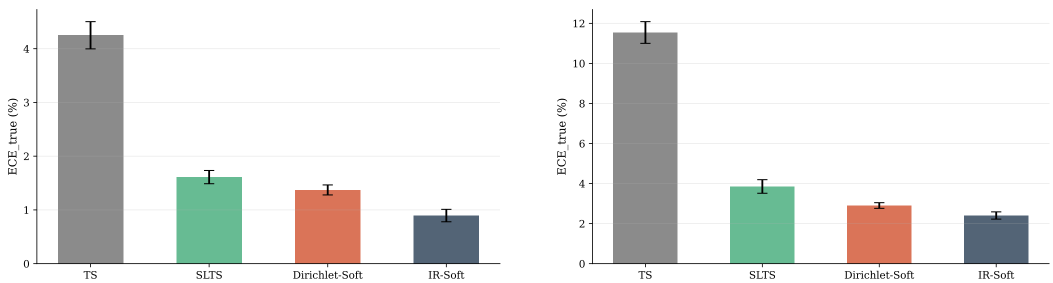

Figure 5: **Multi-seed bar charts.** Mean $\mathrm{ECE}_{\mathrm{true}}$ with error bars ($\pm$std) over 5 seeds. Left: CIFAR-10H ResNet-50. Right: ChaosNLI RoBERTa-Large.

H Reliability Diagrams

Four representative methods are shown in Figure 3 in the main text for CIFAR-10H. Here we provide the corresponding four-method summary for ChaosNLI (RoBERTa-Large), followed by full 3×5 panels covering all 13 methods on both datasets, grouped by type: voted-label baselines (top row), annotation-free and temperature-based soft methods (middle row), and non-parametric/matrix soft methods (bottom row).

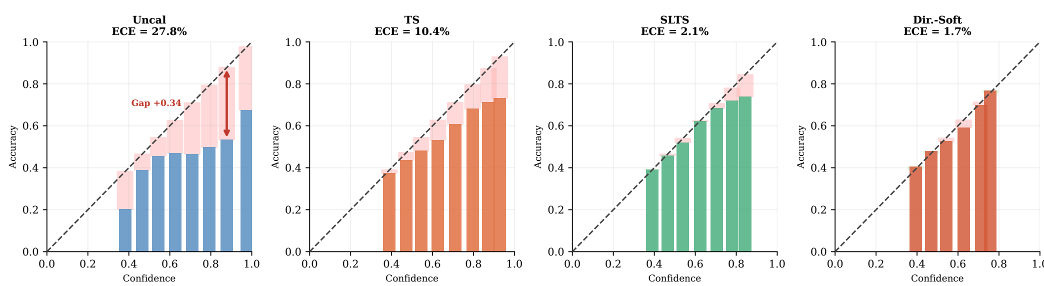

Figure 6: **Reliability diagrams for four representative methods — ChaosNLI RoBERTa-Large ($\mathbf{ECE_{true}}$).** Same method selection as Figure 3. Red shading indicates overconfidence; green indicates underconfidence. NLI models are severely overconfident before calibration; ambiguity-aware methods substantially correct this.

The diagrams confirm the quantitative results in Tables 1–2: TS reduces overconfidence relative to *Uncal* but leaves a large residual gap because it targets voted labels. Dirichlet-Hard performs similarly to TS and sometimes worse, confirming that the problem lies in the target, not model capacity. All soft methods substantially close the gap, with IR-Soft and Dir.-Soft achieving the most uniform residuals.

I DermaMNIST: Supplementary Skin Disease Experiment

See Section 7.4 and Table 2 in the main text.

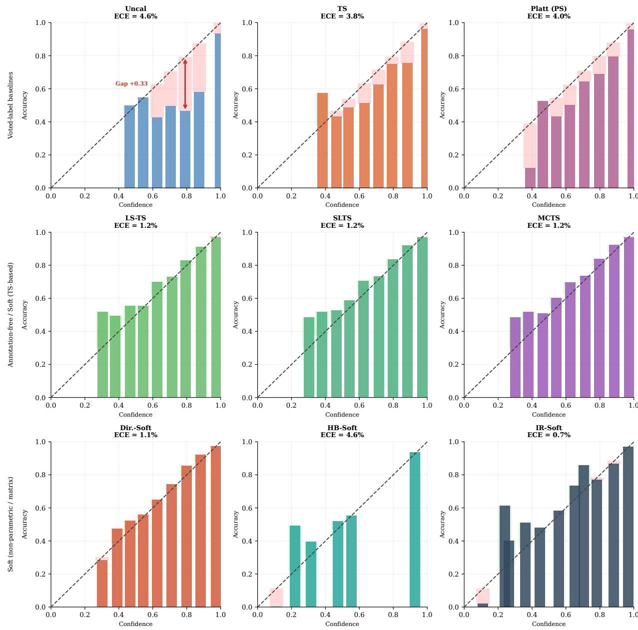

Figure 7: **Reliability diagrams — all methods — CIFAR-10H ResNet-50 ($\mathbf{ECE_{true}}$).** Red shading = overconfident; green = underconfident. Top row: voted-label baselines (Uncal, TS, Platt, Dir.-Hard, HB-Hard). Middle row: annotation-free (LS-TS) and temperature-based soft methods (SLTS, MCTS, SoftPlatt, VS). Bottom row: non-parametric/matrix soft methods (Dir.-Soft, HB-Soft, IR-Soft). Baselines are substantially overconfident; soft methods align closely with the diagonal.

J ISIC 2019 Annotator Confusion Matrix

Construction

Because the full per-image reader annotations of Liu et al. [2020] are not publicly available, we construct a synthetic annotator model using a clinically calibrated 8×8 confusion matrix C, where entry C_{ij} gives the probability that a board-certified dermatologist labels an image as class j when the consensus (majority-vote) diagnosis is class i.

Calibration sources. Diagonal entries (per-condition agreement rates) are set to match the per-condition inter-reader agreement reported in Liu et al. [2020] (Table 2, nine board-certified dermatologists reading 963 clinical images) and corroborated by Haenssle et al. [2018]. Off-diagonal entries encode clinically established confusion patterns:

- **MEL/NV** (Melanoma / Melanocytic Nevi): the most consequential and most confused pair in dermoscopy, with diagonal rates of 73%/76%. Off-diagonal entries $C_{\text{MEL,NV}} = 0.14$ and $C_{\text{NV,MEL}} = 0.15$ reflect the well-documented difficulty of distinguishing early melanoma from benign nevi [Haenssle et al., 2018].
- **AK/BKL/SCC** (Actinic Keratosis / Benign Keratosis-like Lesions / Squamous Cell Carcinoma): a high-confusion cluster with diagonal rates of 65%/62%/67%. The AK$\leftrightarrow$SCC confusion ($C_{\text{AK,SCC}} = 0.16$, $C_{\text{SCC,AK}} = 0.18$) is clinically significant because untreated AK can progress to SCC. BKL shares morphological features with both conditions.
- **DF/VL** (Dermatofibroma / Vascular Lesion): the easiest to distinguish, with diagonal rates of 87%/91%.

The mean diagonal agreement is $\bar{C}_{ii} = 75\%$, matching the aggregate inter-reader agreement reported by Liu et al. [2020] exactly.

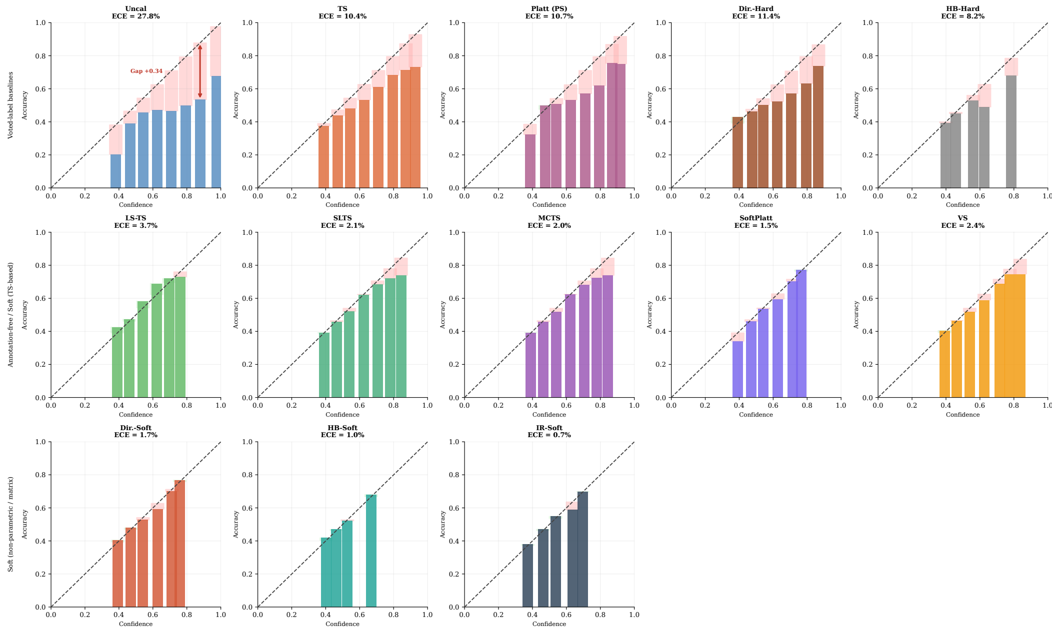

Figure 8: **Reliability diagrams — all methods — ChaosNLI RoBERTa-Large ($\mathbf{ECE_{true}}$).** Same layout as Figure 7. NLI models are severely overconfident before calibration; ambiguity-aware methods substantially correct this.

Simulation procedure. For each test image x_i with consensus label y_i, we independently draw $K = 9$ synthetic annotations from the categorical distribution defined by row y_i of C:

$$a_i^{(k)} \sim \text{Categorical}(C_{y_i,\cdot}), \quad k = 1, \dots, K,$$

and set the empirical label distribution to: $\pi(x_i) = \frac{1}{K}\sum_{k=1}^{K} \mathbf{e}_{a_i^{(k)}}$. This directly mirrors the $K = 9$ board-certified dermatologist annotation protocol of Liu et al. [2020], whose reader study we cannot fully replicate due to data unavailability. The resulting mean annotation entropy over the ISIC 2019 test set is $\bar{H} = 0.66$, reflecting the genuine diagnostic difficulty of the 8-class task.

Visualisation

Figure 9 shows the confusion matrix as a heatmap. The MEL/NV cluster (orange dashed box) and the AK/BKL/SCC high-confusion cluster (purple dotted highlights) are visually prominent. DF and VL have near-perfect diagonal entries and negligible off-diagonal mass.

Full confusion matrix

Table 5 lists the exact numerical entries of the confusion matrix.

K Extended Results

See Tables 1–2 in the main text.

L Implementation Details

CIFAR-10H models. *ResNet-50*: ImageNet-1k weights (`ResNet50_Weights.IMAGENET1K_V1`); FC layer replaced with 2048×10 linear; AdamW, lr$= 10^{-4}$, weight decay 10^{-4}, batch 128, cosine annealing, 30 epochs, 224×224. *ViT-B/16*: ImageNet-21k weights; classification head replaced; AdamW, lr$= 5 \times 10^{-5}$, weight decay 10^{-4}, batch 64, cosine annealing, 20 epochs, 224×224.

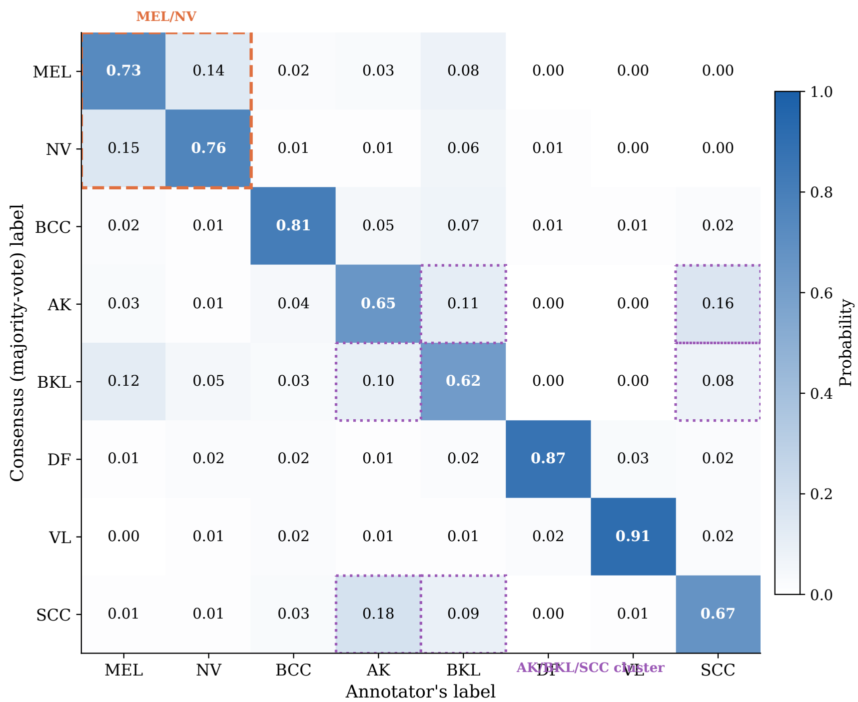

Figure 9: **Inter-reader confusion matrix for ISIC 2019.** Entry C_{ij} is the probability that a dermatologist labels an image as class j given the consensus label is i. Diagonal entries (bold) are the per-condition agreement rates, calibrated to match Liu et al. [2020] Table 2. Orange dashed box: MEL/NV most-confused pair. Purple dotted highlights: AK/BKL/SCC high-confusion cluster. Mean diagonal agreement $\bar{C}_{ii} = 75\%$.

Table 5: **Numerical values of the inter-reader confusion matrix for ISIC 2019.** Row = consensus (majority-vote) label; column = annotator's label. Each row sums to 1. Diagonal entries (bold) are per-condition agreement rates, calibrated to Liu et al. [2020] Table 2; mean diagonal agreement $\bar{C}_{ii} = 75\%$.

	MEL	NV	BCC	AK	BKL	DF	VL	SCC
MEL	**0.73**	0.14	0.02	0.03	0.08	0.00	0.00	0.00
NV	0.15	**0.76**	0.01	0.01	0.06	0.01	0.00	0.00
BCC	0.02	0.01	**0.81**	0.05	0.07	0.01	0.01	0.02
AK	0.03	0.01	0.04	**0.65**	0.11	0.00	0.00	0.16
BKL	0.12	0.05	0.03	0.10	**0.62**	0.00	0.00	0.08
DF	0.01	0.02	0.02	0.01	0.02	**0.87**	0.03	0.02
VL	0.00	0.01	0.02	0.01	0.01	0.02	**0.91**	0.02
SCC	0.01	0.01	0.03	0.18	0.09	0.00	0.01	**0.67**

ChaosNLI models. *RoBERTa-Large*: `roberta-large-mnli` from HuggingFace, pre-fine-tuned on MultiNLI [Liu et al., 2019]; no additional fine-tuning. *DeBERTa-v3-base*: `cross-encoder/nli-deberta-v3-base` from HuggingFace [He et al., 2023]; no additional fine-tuning. Both models use the combined ChaosNLI SNLI+MNLI subset ($n = 3{,}113$), split 50/50 into calibration and test (stratified, seed 42).

ISIC 2019 models. *EfficientNet-B4*: ImageNet-1k weights; stratified 70/15/15 split; class-weighted cross-entropy (NV: $\approx 67\%$ of training images); AdamW, lr$= 3 \times 10^{-4}$, weight decay 10^{-4}, 2-epoch linear warm-up + cosine decay, 20 epochs, 380×380. *ViT-S/16*: ImageNet-21k weights; same split and loss weighting; AdamW, lr$= 10^{-4}$, weight decay 10^{-4}, 2-epoch warm-up + cosine decay, 20 epochs, 224×224.

DermaMNIST models. *ResNet-18*: ImageNet-1k weights, class-weighted cross-entropy, AdamW, lr$=10^{-4}$, 30 epochs. *ViT-S/16*: ImageNet-21k weights, same training protocol, 224×224.

Calibration optimizers. TS, SLTS: LBFGS, lr$=0.1$, 500 iterations, tolerance 10^{-9} (gradient), 10^{-11} (loss). Platt/VS: Adam, lr$=0.01/0.05$, weight decay 10^{-4}, 2000 steps.

ECE bins. $B = 15$ equal-width bins in $[0, 1]$; bins with fewer than 1 example are excluded.

Compute. CIFAR-10H fine-tuning: ≈20 min on one NVIDIA A100. All calibration methods: < 60 s on CPU after logit caching.